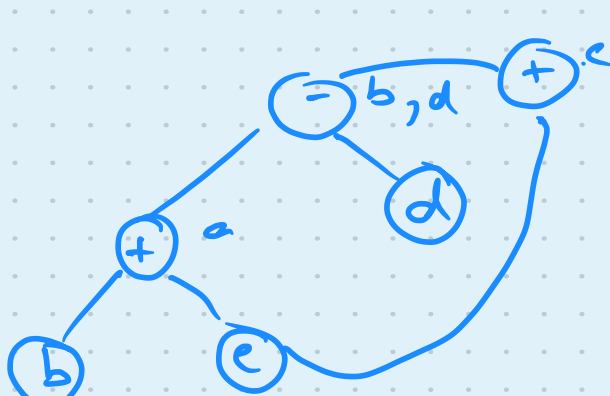


(optimisation of control flow graph)

LOCAL OPTIMISATION USING DAG

$$\begin{aligned} a &= b + c \\ b &= a - d \\ c &= b + c \\ d &= a - d \end{aligned}$$


Algorithm..

1. $a = a - b$
2. $u = a - c$
3. $v = t + u$
4. $d = v + u$

a	b	c	d	u	v	t
T	T	F	F	F	F	F
T	F	T	F	F	F	F
F	F	F	F	T	F	T
F	F	F	F	T	T	F
F	F	F	F	F	F	F

LIVENESS

Variable x is live at statement s if:

- There exists s' that uses x
- There is a path from s to s'
- That path has no intervening assignment to x

Definitions

- ① use_B : set of vars used in B prior to any definition of the var
- ② def_B : set of vars used in B prior to any use of var in B
- ③ $IN[B]$: set of vars live on entry to B
- ④ $OUT[B]$: set of vars live on exit from B

Algorithm

$IN[EXIT] = \emptyset$ } no variables live on exit from pgm

for (each B other than EXIT) $IN[B] = \phi$;

while (changes to any IN occur)

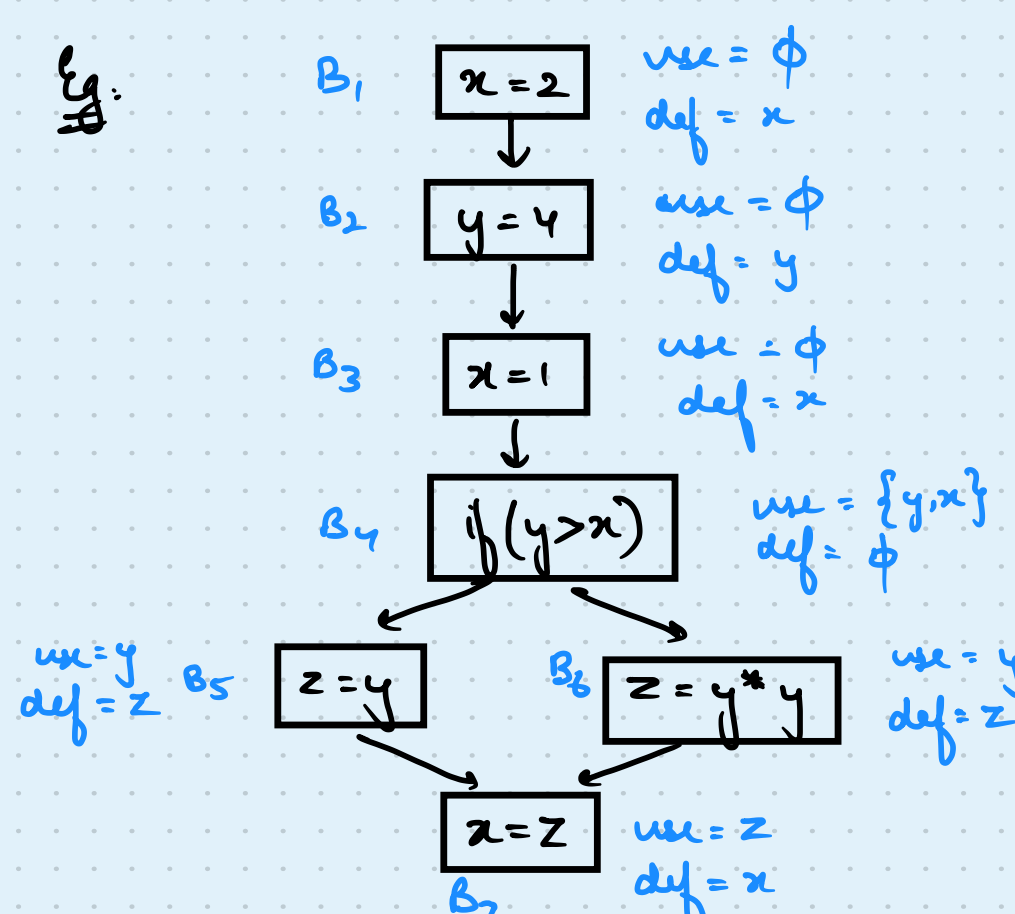
for (each B other than EXIT) {

$$\text{OUT}[B] = \bigcup_{s \in S} \text{IN}[s]$$

all successors of B

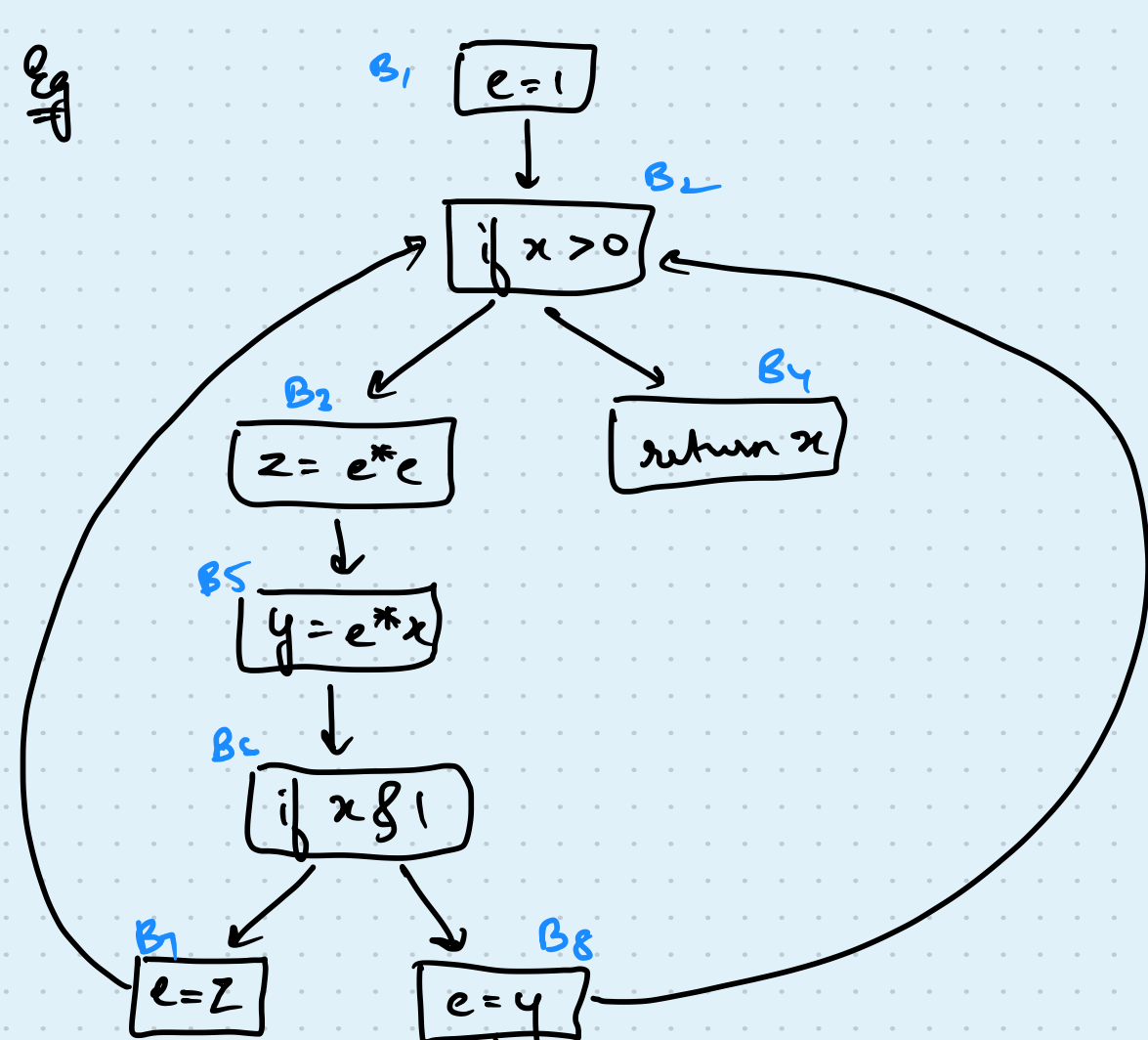
$$IN[B] = use_B \cup (OUT[B] - def_B)$$

9



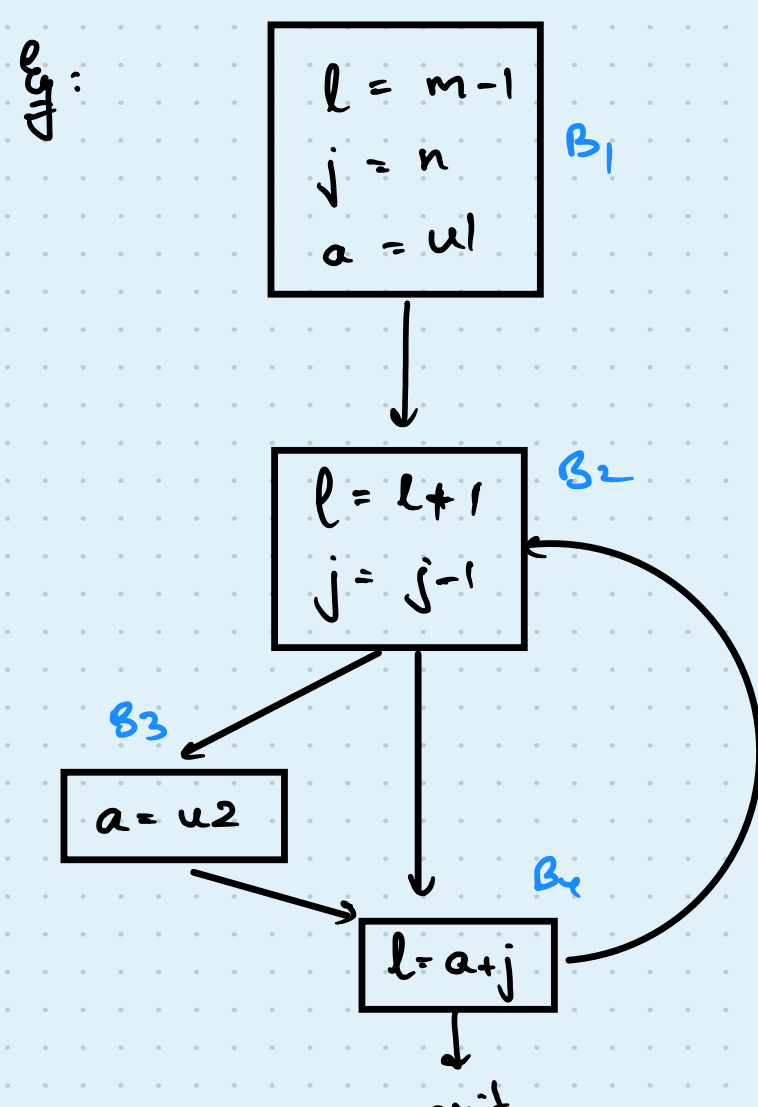
Applying algorithm:

Block	IN	OUT
B ₁	ϕ	
B ₂	ϕ	
B ₃	ϕ y	y {y, x}
B ₄	ϕ {y, x}	y
B ₅	ϕ y	z
B ₆	ϕ y	z
B ₇	ϕ z	ϕ

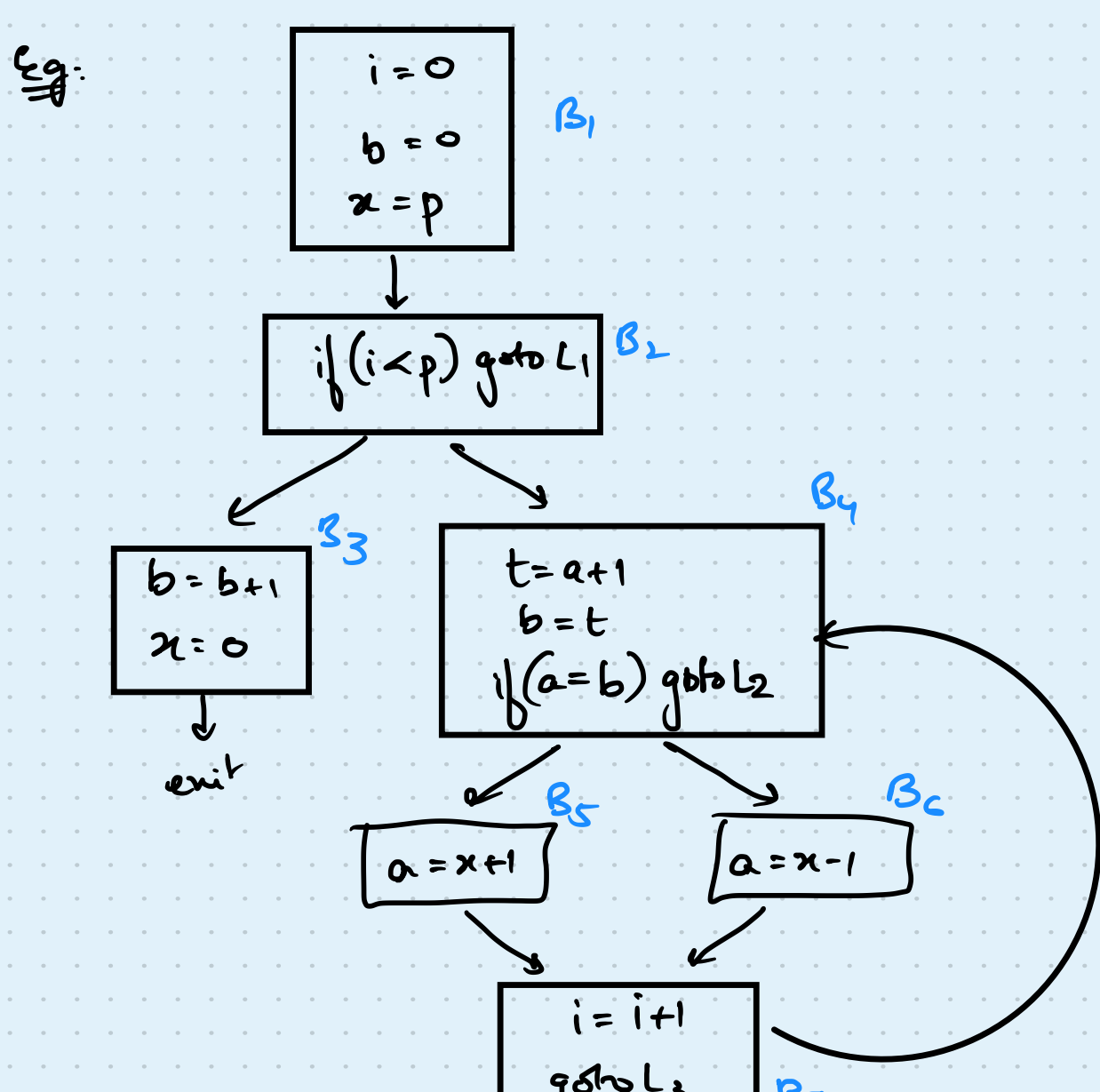


Block	use	def
B_1	ϕ	e
B_2	x	ϕ
B_3	e	z
B_4	x	ϕ
B_5	$\{e, x\}$	y
B_6	x	ϕ
B_7	z	e
B_8	y	e

Block	IN	OUT
B ₁	$\phi \{e\}$	$\{e, n\}$
B ₂	$\phi \{e, n\}$	$\{e, n\}$
B ₃	$\phi \{e, n\}$	$\{e, n, z\}$
B ₄	$\phi \{n\}$	ϕ
B ₅	$\phi \{e, n, z\}$	$\{n, y, z\}$
B ₆	$\phi \{n, y, z\}$	$\{z, y\} \{n, y, z\}$
B ₇	$\phi \neq \{n, z\}$	$\phi \{e, n\}$
B ₈	$\phi \neq \{y, z\}$	$\phi \{e, n\}$



Block	use	def	IN	OUT
B_1	$\{m, n, u\}$	$\{l, j, a\}$	$\emptyset$ $\{m, n, u, u_2\}$	$\{l, j, a, u_2\}$
B_2	$\{l, j\}$	$\{l, j\}$	$\emptyset$ $\{l, j, a, u_2\}$	$\{a, j, u_2\}$
B_3	$\{u_2\}$	$\{a\}$	$\emptyset$ $\{u_2, j\}$	$\{a, j\}$ $\{a, j, u_2\}$
B_4	$\{a, j\}$	$\{l\}$	$\emptyset$ $\{a, j\}$ $\{a, j, u_2\}$	$\emptyset$ $\{l, j, a, u_2\}$



Block	use	def
B_1	p	i, b, x
B_2	i, p	ϕ
B_3	b	x
B_4	a	t, b
B_5	x	a
B_6	x	a
B_7	i	ϕ

Block	IN	OUT
B ₁	∅ {a, p}	{a, b, x, i, p}
B ₂	∅ {a, b, x, i, p}	{a, b, x, i}
B ₃	∅ b	∅
B ₄	∅ {a, x, i}	<u>{x, i}</u>
B ₅	∅ <u>{x, i}</u>	✓ {a, x, i}
B ₆	∅ <u>{x, i}</u>	✓ {a, x, i}
B ₇	∅ ✓ {a, x, i}	∅ {a, x, i}